CHARACTERIZATIONS OF SHEAF THEORETICALLY EMPTY OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The plus construction and its relation to sheafification. Same chapter, Theorem theorem-plus. Shows that the plus construction applied to the empty presheaf is its sheafification.
- The sheaf condition for the empty covering. Same chapter, Remark remark-sheaf-condition-empty-covering. Supplies the implication from the empty family being a covering to values being singletons.
- The axioms of a site. Same chapter, Definition definition-site. Supplies the stability axiom used for the final assertion.

Source reference. Original theorem 00WB: sources/stacks-sites/pinned-original-source.tex, source lines 10158–10158. Navigation pointer invoked at proof line 10158; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original remark 04B3: sources/stacks-sites/pinned-original-source.tex, source lines 10162–10162. Navigation pointer invoked at proof line 10162; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original definition 00VH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 526–526. Navigation pointer invoked at proof line 10165; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let U be an object of $\mathcal{C}$. The following are equivalent:*

(1) *U is sheaf theoretically empty,*
(2) *$\mathcal{F}(U)$ is a singleton for each sheaf $\mathcal{F}$,*
(3) *$\emptyset^\#(U)$ is a singleton,*
(4) *$\emptyset^\#(U)$ is nonempty, and*
(5) *the empty family is a covering of U in $\mathcal{C}$.*

Moreover, if U is sheaf theoretically empty, then for any morphism $U' \to U$ of $\mathcal{C}$ the object U' is sheaf theoretically empty.

Proof. For any sheaf $\mathcal{F}$ we have $\mathcal{F}(U) = \mathrm{Mor}_{Sh(\mathcal{C})}(h_U^\#, \mathcal{F})$. Hence, we see that (1) and (2) are equivalent. It is clear that (2) implies (3) implies (4). If every covering of U is given by a nonempty family, then $\emptyset^+(U)$ is empty by definition of the plus construction. Note that $\emptyset^+ = \emptyset^\#$ as $\emptyset$ is a separated presheaf, see Theorem 00WB. Thus we see that (4) implies (5). If (5) holds, then $\mathcal{F}(U)$ is a singleton for every sheaf $\mathcal{F}$ by the sheaf condition for $\mathcal{F}$, see Remark 04B3. Thus (5) implies (2) and (1) – (5) are equivalent. The final assertion of the lemma follows from Axiom (3) of Definition 00VH applied the empty covering of U. □